

Deep Learning Approaches for Exoplanet Detection using the Transit Method on Light Curve Data

Syeda Saarah Fatema¹

Department of Information Science and engineering
Ramaiah Institute of Technology
Bengaluru, India
syedasaarahfatema@gmail.com

S R Mani Sekhar²

Department of Information Science and engineering
Ramaiah Institute of Technology
Bengaluru, India
manisekharsr@msrit.edu

Abstract— Exoplanet detection has appeared as one of the most interesting edges in modern astronomy, giving meaningful insights about planetary systems that are beyond our imagination. With over 5,000 confirmed exoplanets discovered to date, there are many detection techniques that includes radial velocity, direct imaging, and gravitational microlensing have remarkable contribution to this field. Even though there are several methods to detect the exoplanets, transit method is exceptional for its ability to provide information about a planet size, orbital period, and potential habitability. The Transiting Exoplanet Survey Satellite telescope has played a vital role in the field of astronomy by collecting huge amount of light curve data from millions of stars, understanding that life exists beyond earth. Separating true planetary transits from noisy and other stellar variations within this data remains a challenging and resource-intensive task. To enhance the detection accuracy deep learning models such as Feedforward Neural Networks, Bayesian Neural Networks, and Recurrent Neural Networks are trained to understand the patterns in light curve data. These models have detection accuracies of 95%, 91%, and 96%, respectively. By leveraging the uniqueness of each model FNN for structured data, BNN for uncertainty estimation, and RNN work better with time-series data, these methods work together effectively to classify the planetary transit events. This research aims to enhance the exoplanet discovery, enhance classification accuracy, and reduce computational costs when compare to object detection models and reduce time consumption.

Keywords—TESS (Transiting Exoplanet Survey Satellite), FNN (Feedforward Neural Networks), BNN (Bayesian Neural Networks), Exoplanet, RNN(Recurrent Neural Networks), Transit method, deep learning.

I. INTRODUCTION

Detecting exoplanets is an important view of increasing our knowledge of the universe. Through these studies, scientists have investigated the conditions that would allow these planets to support life based on their physical features and composition. Therefore, knowledge of exoplanets provides deep insights into the processes behind planetary formation, the complex dynamics within planetary systems, and the potential environments that may support life on other worlds.

Understand that life exists beyond the earth and there are vast number of planets located in universe like exoplanets, exomoons, Exostar's that are located outside our solar system, this has been a great interest for astronomers since the first discovery of exoplanet was done. There exists thousands and

millions of planetary systems that exist in our galaxy[1][2]. These findings are essential to understand how planets form, evolve, and potentially support life. The detection of exoplanets in the "habitable zone" of their stars, where life would exist presents a probability that there could be other planets that support life.

Numerous methods have been discovered to identify exoplanets, each contributing unique strengths and challenges. Admist these, the transit method is one of the easier techniques to identify exoplanets. It detects dimming light in a star's brightness, whenever a planet passes through the star, the brightness is reduced, then its concluded that there exists an exoplanet. Another approach that is commonly used is the radial velocity method, which observes slight shifts in a star's motion caused by the gravitational pull of an orbiting planet, resulting in a subtle wobble. Direct imaging seeks to capture images of exoplanets by minimizing or blocking the intense light emitted by their host stars, detecting exoplanet becomes a challenge with this technique. Gravitational microlensing detects exoplanets by observing the bending of light from a more distant star it is similar to direct imaging. Each of these techniques plays an important role in the effort to discover and study exoplanets across the universe. Admist these methods, the transit has become one of the most well-to-do and widely used techniques to discover exoplanets. It works by detecting slight dips in a star's brightness when a planet passes in front of it, blocking some of the star's light. The timing, frequency, and depth of these dimming events provide sensitive information about the planet size, orbital period and other features. TESS telescope was setup in 2018, has changed the detection of exoplanets using the transit method. TESS is intended to observe large number of stars across the universe, collecting high quality data. This allows astronomers to uncover exoplanets near the stars and discover a wide variety of planetary systems. With its high precision light curve data, TESS has huge contribution to the detection of small exoplanets, specifically for those that might be similar in size to Earth. Recent studies in machine learning and deep learning offers powerful solutions for automating the detection of exoplanets through the transit method [3]. Models such as FNN and BNN are good for analysing complex datasets like light curve data. The models are capable of recognizing patterns, separating real planet transits from noisy ones with better accuracy. RNN is ideal to work with processing time-series data, which is ideal for recognizing the nature of exoplanet transits. While the transit method remains one of the most efficient method for exoplanet detection[4], it has some drawbacks. Not all exoplanets will have orbits that align in such a way that they transit aligns with their host stars.

This means the transit method is biased towards capturing only those planets that pass directly in front of their star. However, smaller planets, such as Earth-sized worlds, produce subtle transits, which makes them hard to detect, especially when their orbits are far from the star. Irrespective of these drawbacks, the transit method continues to be a useful technique in the search for exoplanets. It has led to the discovery of thousands of exoplanets, many of which have been studied in detail. Data from missions like Kepler and TESS telescope have greatly expanded our knowledge of planetary and solar systems, particularly in the search for planets like earth that may exist in the habitable zone of their stars. Applying deep learning techniques and machine learning techniques to the huge datasets collected by the telescope that offers a beneficial approach to enhance exoplanet detection. These advanced models can efficiently automate the identification of planetary transits, minimizing the reliance on manual analysis and increases speeding up the discovery process.

II. RELATED WORK

YOLO provides faster and accurate detection when compared to other traditional machine learning object detection models. The YOLOv5 model demonstrated significant performance on TESS data with mean average accuracy of 0.82, they are trained on 221 annotated light curves from the TESS telescope [5]. Detection accuracy of SVM model by removing outliers from the dataset. Simultaneously, the kNN algorithm, trained on Kepler data, successfully classified exoplanets and K-Means clustering method grouped planets according to their traits, uncovering patterns such as the higher frequency of Hot Jupiter's with shorter orbital periods [6].

To address this, the study [7] introduces deep PACO a hybrid approach that integrates PACO statistical noise modeling with a Convolutional Neural Network for enhanced signal detection and mapping. This method significantly improves detection contrast, particularly beyond 0.5 arcseconds, offering up to half a magnitude better performance compared to PACO alone. Validated using metrics such as F1 Score, True Positive Rate, and False Discovery Rate, deep PACO delivers high accuracy and reduced false positives, establishing a new benchmark for exoplanet imaging techniques.[8] presents constructs, a deep-learning algorithm for direct exoplanet detection in high-contrast images, designed to identify faint signals obscured by stellar speckle noise. Using a convolutional autoencoder trained on a reference library and enhanced with ridge regression, Construct improves sensitivity in angular differential imaging sequences. Comparative tests show it outperforms PCA, achieving higher signal-to-noise ratios in 67% of cases and enhancing contrast by up to 2.6 times. These results demonstrate Construct's potential for advancing high-contrast imaging. Research [9] demonstrates machine learning-based approach for detecting exoplanets by analyzing light curves, which represent variations in a star's brightness over time. Researchers used TSFRESH to extract 789 key features from each light curve, capturing essential patterns associated with potential planetary transits. These features were then fed into a LightGBM gradient boosting classifier for training. When tested on NASA's K2 mission data with simulated planet signals, the method achieved impressive accuracy—96% on Kepler data and 98% on TESS data. This approach offers a faster and more resource-efficient alternative to traditional techniques, positioning it as a highly effective tool for advancing exoplanet discovery. Where [10] explores five ensemble machine learning algorithms used are Adaboost, Random Forest, Stacking, Random Subspace

Method and Extremely Randomized trees to enhance exoplanet detection with the Kepler Object of Interest dataset. Each model was fine-tuned using cross-validation and evaluated on accuracy, precision, sensitivity, and F1 score. The Stacking algorithm emerged as the top performer, benefiting from its ability to integrate predictions from multiple models achieving over 80% accuracy, this study highlights the effectiveness of ensemble methods in improving exoplanet classification. Where [11] introduces a CNN-based model for detecting exoplanets by identifying planetary transits in Kepler light curves. The model is trained on a combination of real and synthetic data, with synthetic curves generated using a limb-darkening models.

Using genetic algorithms and sensitivity analysis, the optimal data balance is determined. Phase-folding transforms light curves into images for CNN classification. The model achieved 98% accuracy, 97.05% precision, and a 99% true positive rate, surpassing previous methods and improving transit detection in real data. To tackle the challenges of enhancing exoplanet detection [12] removes noise from stellar activity in radial velocity (RV) data, the limitation in achieving precise measurements. Without depending on time-series data, machine learning techniques such as neural networks and regression are used to detect and eliminate unwanted noisy signals. They used simulated RV data from SOAP 2.0 to train their algorithms. Using AI, notably CNNs, which are efficient at processing massive volumes of telescope data and overcoming obstacles in conventional detection approaches, the research [13] aims to improve the discovery of exoplanets. When employed to detect starlight patterns that suggest the existence of exoplanets, CNNs demonstrated a high degree of accuracy in differentiating real signals from noise. The CNN model showed a promising increase in accuracy and efficiency over previous techniques, with a notable detection accuracy of 98%. This artificial intelligence-driven strategy is opening the door for advances in exoplanetary exploration and discovery by increasingly confirmed exoplanet detections and providing meaningful insights into the habitable worlds. Enhancing exoplanet detection accuracy using statistical techniques applied to radial velocity data. Bayesian models and Gaussian processes to interpret RV signals within noise from stellar activity, researchers achieved an improved sensitivity to planetary signals, with detection rates showing remarkable improvement. Results shown that these methods reduce false positives and increase confidence in detecting smaller, Earth-like planets, achieving up to a 20% improvement in signal accuracy. This refined approach provides a robust foundation for future space missions aimed at identifying and characterizing potentially habitable exoplanets [14]. This research [15] improves the signal-to-noise ratio for faint signals, reduces the noise in the data, achieving better accuracy and precision.

III. PROPOSED METHODOLOGY

The detection of exoplanets addresses significant challenges in the field of astronomy, requires effective and efficient methods to accurately identify planetary transits within from the complex datasets. This research presents a neural network-based approach for exoplanet detection using TESS (Transiting Exoplanet Survey Satellite) mission data. It addresses gaps in existing research, such as reducing computational costs, handling false positives and aims flexibility across various exoplanet detection techniques. By separately assessing the performance of BNN, RNN, FNN.

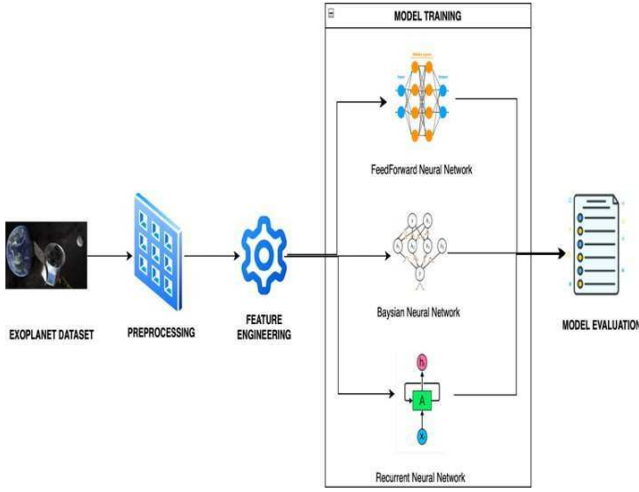


Fig.1. Architecture design for working of exoplanet detection

As shown in Fig.1. data consists of time-series star brightness measurements, with dips in brightness potentially signaling planetary transits and other important features. The first step involves data cleaning which addresses missing values through imputation. To ensure that the features are on a comparable scale, z-score normalization is applied. The next step is to transform into a structured format for input into deep learning models. Feature engineering for exoplanet detection involves finding relevant features from TESS light curves to improve model accuracy. Important features include statistical measures like mean, median, variance, and standard deviation to represent the star's overall brightness behavior. In addition, the Imbalance of the data is calculated to describe the shape of the light curve's distribution. These features helps in identifying patterns related with planet transits.

As shown in [16] Feedforward Neural Network model is trained to detect exoplanets from telescope light curve data. It addresses class imbalance using ADASYN resampling method is applied to enhance the presentation of the minority class. The FNN architecture contains dense layers, dropout, and batch normalization, with L2 regularization which reduces overfitting. During training the data early stopping and learning rate reduction callbacks are implemented to finetune the model. Bayesian Neural Network model [17] uses the Monte Carlo (MC) Dropout method for exoplanet detection. The BNN integrates dense layers, dropout layers, and batch normalization to enhance model performance and reduce overfitting. MC Dropout is applied during both training and testing, which allows the model to have a precise distribution over the network weights. This approach helps capture the uncertainty in the predictions by evaluating dropout during training time, adding different stochastic networks. By using MC Dropout, the model can handle uncertainty or noisy data, which is often present in astronomical datasets like TESS light curves. The model benefits from the uncertainty estimation, improving robustness and reducing overfitting by preventing the model from become reliable on specific weights or features. The RNN model, utilizes LSTM layers, plays an important role in the exoplanet detection task by efficiently managing sequential data, such as the time-series TESS light curves. These light curves, which represent variations in star brightness over time, contains temporal patterns that standard machine learning models may struggle to capture [18]. The LSTM layers within the RNN aims to learn the time series relationship, helps the model to recognize regular changes due to exoplanet transits. This

capability allows the model to make precise predictions about exoplanets, even when faced with noisy or incomplete data. By leveraging RNNs, the model gains better understanding of the star's behavior, eventually adding in the more efficient and accurate identification of exoplanets. Accuracy measures how often a model correctly predicts output by calculating the ratio of correct predictions to the total number of predictions, though it may be unreliable in imbalanced datasets. Precision indicates the percentage of correctly predicted positive instances among all predicted positives, while recall measures how well the model captures all actual positives. RNN attains the higher accuracy of 96.01%, surpasses FNN slightly, making it the best overall model among these models. Even though BNN shows better recall for class 1, its lower overall accuracy and precision make it less suitable for this task compared to RNN or FNN.

Algorithm 1 FNN-based Exoplanet Detection

Input: TESS Dataset $D = \{(x_i, y_i) \mid i = 1, 2, \dots, N\}$, $x_i \in \mathbb{R}^d$, $y_i \in \{0, 1\}$

Output: Trained FNN Model M , Accuracy Score A

- 1: Initialize weights $W^{(l)}$ and biases $b^{(l)}$ using He/Xavier method
- 2: **for** each epoch **do**
- 3: **for** each training sample (x_i, y_i) **do**
- 4: Compute hidden layer activations using:

$$a^{(l)} = \max(0, W^{(l)}a^{(l-1)} + b^{(l)})$$
- 5: Apply dropout:

$$a^{(l)} = a^{(l)} \cdot \mathbb{I}(u > p), \quad u \sim \mathcal{U}(0, 1)$$
- 6: Compute output:

$$\hat{y}_i = \sigma(W^{(L)}a^{(L-1)} + b^{(L)})$$
- 7: Compute binary cross-entropy loss:

$$L = -[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$
- 8: Update weights using Adam optimizer:

$$\theta = \theta - \eta \cdot \nabla_{\theta} L$$
- 9: **end for**
- 10: **end for**
- 11: Compute accuracy:

$$A = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\hat{y}_i = y_i)$$

- 12: **return** M, A
-

IV. RESULTS AND DISCUSSION

TESS dataset consist of 39,000 instances and more than 200 features that are collected from NASA's website. The dataset required careful preprocessing, imputation, feature selection, class imbalance was handles using SMOTE. The proposed work analyzes the performance of the FNN, BNN, and RNN models using key evaluation metrics, including accuracy, precision, recall. The discussion focuses on identifying each model's strengths and weaknesses, with particular attention to their ability to handle an imbalanced dataset. While RNN achieved the highest overall accuracy, BNN demonstrated superior recall for the minority class, albeit with lower precision. The following analysis offers a detailed comparison of the models, examines their implications, and evaluates their suitability for similar classification challenge.

Table1 : Results of Feedforward Neural Network , Bayesian Neural Network and Recurrent Neural Network models.

MODEL	Accuracy in percentage (%)	class	Recal l	F1-score
FNN (Feed forward Neural Network)	95	0	0.96	0.98
		1	0.20	0.03
BNN (Bayesian Neural Network)	91	0	0.91	0.95
		1	0.40	0.30
RNN (Recurrent Neural Network)	96	0	0.96	0.98
		1	0.20	0.03

As shown in Fig.2 and Fig.3 graphs show the training and validation performance of a Feedforward Neural Network over 50epochs. Training accuracy plateaus around 95%, while validation accuracy stabilizes near 92%, with slight fluctuations. Both training and validation loss decrease consistently, showing effective convergence with minimal overfitting. For class 0 recall and precision is 0.96 and 0.98 and for class1 it is 0.20 and 0.03 respectively.

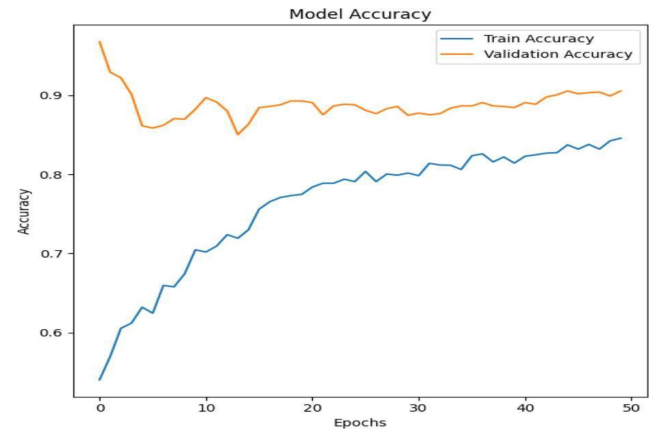


Fig.4. Model Accuracy of BNN for 50 epochs

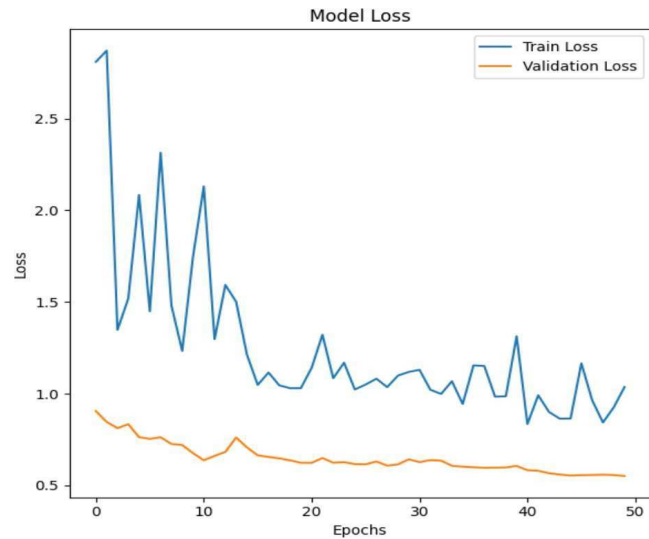


Fig.5. Model loss of BNN for 50 epochs

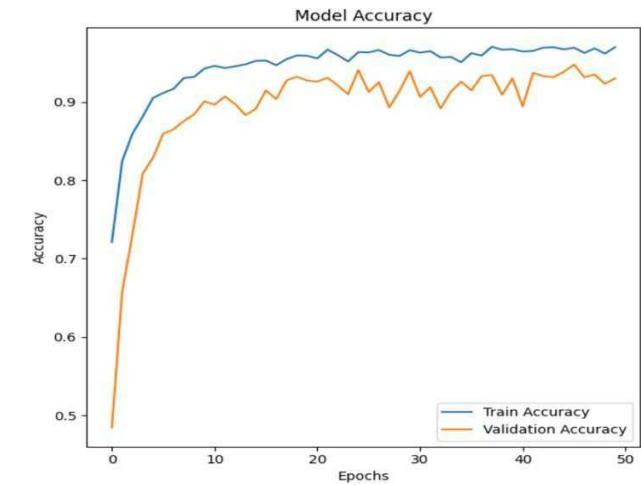


Fig.2. Model Accuracy of FNN for 50 epochs

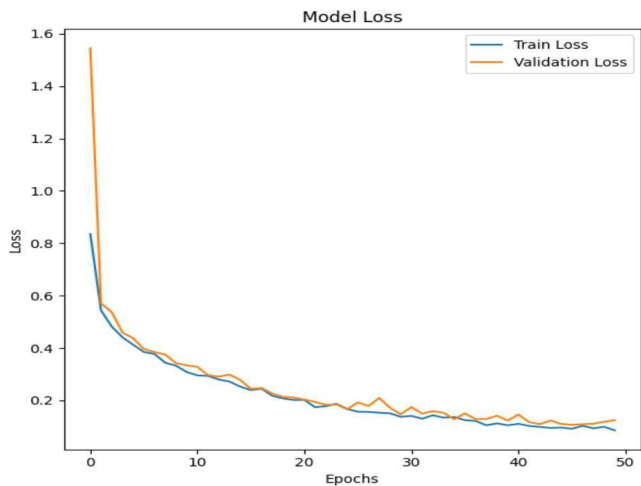


Fig.3. Model loss of FNN for 50 epochs

As shown in Fig.4 and Fig.5 BNN model maintains consistently high validation accuracy 91% from the beginning, while training accuracy steadily increases to around 85% over time. Although the validation loss stays low and stable, the training loss exhibits significant fluctuations, indicating unstable convergence. Despite this instability, the model demonstrates strong generalization performance on the validation data. For class 0 recall and precision is 0.91 and 0.95 and for class1 it is 0.40 and 0.30 respectively.

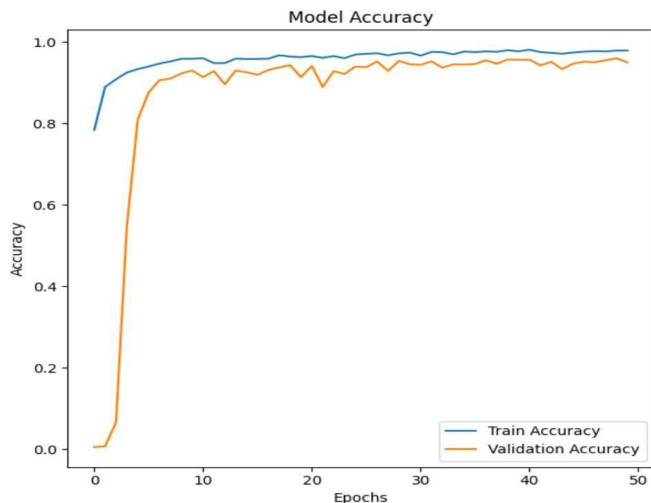


Fig.5. Model Accuracy of RNN for 50 epochs

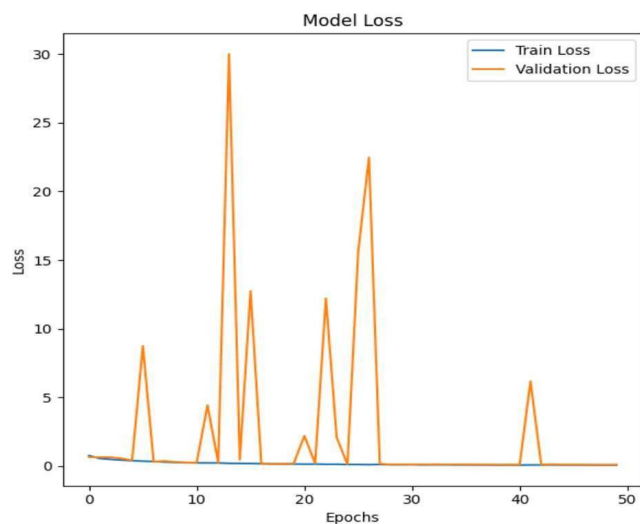


Fig.6. Model Accuracy of RNN for 50 epochs

As demonstrated in Fig.5 and Fig.6 accuracy graph shows a steady increase in training accuracy, eventually stabilizing near 96%, suggesting that the model effectively learns the training data. The loss graph reveals that the training loss stays consistently low and stable, demonstrating a strong fit to the training data. Using the evaluation metrics such as accuracy, precision and recall this model detects non-exoplanet better than exoplanet. For class 0 recall and precision is 0.96 and 0.98 and for class1 it is 0.20 and 0.03 respectively.

V. CONCLUSION AND FUTURE WORK

This research investigates the use FNN, BNN, and RNN for identifying exoplanets. Among the evaluated models, RNN emerged as the most effective, achieving an accuracy of 96.01% due to its exceptional ability to capture temporal patterns in time-series data. While BNN excelled in recall for minority classes and demonstrated strong uncertainty estimation, it experienced instability during convergence and lower overall precision. On the other hand, FNN displayed solid generalization capabilities but lacked the proficiency of RNN in managing sequential dependencies.

Future enhancements, the study proposes integrating hybrid models that leverage the strengths of all three neural network architectures to enhance detection capabilities. Improved data preprocessing strategies are recommended to better manage noise and refine light curve clarity. The adoption of Explainable AI (XAI) methods is emphasized to ensure greater transparency and interpretability of model predictions.

References

- [1]. Raminaidu, Ch, et al. "Building Accurate Machine Learning Models for Predicting the Habitability of Exo-Planets." *2023 5th International Conference on Smart Systems and Inventive Technology (ICSSIT)*. IEEE, 2023.
- [2]. Al-Mamun, A. M., M. R. Hossain, and M. M. Sharmin. "Exoplanets detection using lite convolutional neural networks (LCNN)." *Material Sci & Eng 7.4* (2023).
- [3]. Wang, Kaitlyn, et al. "The GPU phase folding and deep learning method for detecting exoplanet transits." *Monthly Notices of the Royal Astronomical Society* 528.3 (2024).
- [4]. Singh, Jobanbir, and Nongmeikapam Thoiba Singh. "Analysis of Exoplanet Detection Methods using Machine Learning and Deep Neural Networks." *2023 International Conference on Sustainable Computing and Smart Systems (ICSCSS)*. IEEE, 2023.
- [5]. Sekhar, S. M., Tejas, C., Kanna, V. N., & Kaur, A. (2023). Finding exoplanets using object detection. *Astrophysics and Space Science*
- [6]. Herur, Aahish Nagesh, Raquib Tajmohamed, and J. Godwin Ponsam. "Detection and classification of exoplanets using hybrid kNN model." *2022 International Conference on Computer Communication and Informatics (ICCCI)*. IEEE, 2022.
- [7]. Flasseur, O., Bodrito, T., Mairal, J., Ponce, J., Langlois, M., & Lagrange, A. M. (2024). deep PACO: Combining statistical models with deep learning for exoplanet detection and characterization in direct imaging at high contrast. *Monthly Notices of the Royal Astronomical Society*
- [8]. Wolf, Trevor N., Brandon A. Jones, and Brendan P. Bowler. "Direct Exoplanet Detection using Convolutional Image Reconstruction (ConStruct): A New Algorithm for Post-processing High-contrast Images." *The Astronomical Journal* 167.3 (2024)
- [9]. Malik, Abhishek, Benjamin P. Moster, and Christian Obermeier. "Exoplanet detection using machine learning." *Monthly Notices of the Royal Astronomical journal*
- [10]. Luz, Thiago SF, Rodrigo AS Braga, and Enio R. Ribeiro. "Assessment of Ensemble-Based Machine Learning Algorithms for Exoplanet Identification." *Electronics* 13.19 (2024)
- [11]. Cuéllar S, Granados P, Fabregas E, Curé M, Vargas H, Dormido-Canto S, Farias G. Deep learning exoplanets detection by combining real and synthetic data. *PLoS One*. 2022 May 25;17(5):e0268199. doi: 10.1371/journal.
- [12]. Identifying Exoplanets with Deep Learning. IV. Removing Stellar Activity Signals from Radial Velocity Measurements Using Neural Networks ,Zoe. L. de Beurs, Andrew Vanderburg, Christopher J. Shallue, *et al* ,iopscience.
- [13]. Kyle A. Pearson, Leon Palafox, Caitlin A. Griffith, searching for exoplanets using artificial intelligence, *Monthly Notices of the Royal Astronomical Society,journal, academic.oup*.
- [14]. Hara, Nathan C., and Eric B. Ford. "Statistical methods for exoplanet detection with radial velocities." *Annual Review of Statistics and Its Application* 10.1 (2023),journal

- [15]. G. Gopan, M. Benisha and M. Sathiyarayanan, "Chandrayaan-3: Proposed System to Detect and Identify Exoplanets in the Deep Space using AI," *2023 International Conference on new Frontiers in Communication, Automation, Management and Security (ICCAMS)*, Bangalore, India, 2023.
- [16]. Sazli, Murat. (2006). A brief review of feed-forward neural networks. Communications Faculty Of Science University of Ankara
- [17]. Charnock, T., Levasseur, L.P., & Lanusse, F. (2020). Bayesian Neural Networks. *ArXiv, abs/2006.01490*.
- [18]. Das, S., Tariq, A., Santos, T., Kantareddy, S.S., Banerjee, I. (2023). Recurrent Neural Networks (RNNs): Architectures, Training Tricks, and Introduction to Influential Research. In: Colliot, O. (eds) *Machine Learning for Brain Disorders. Neuromethods*, vol 197. Humana, New York.